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FACULTY OF COMPUTING AND MATHEMATICAL SCIENCES
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

PROJECT REPORT ENTITLED

AGRONEXUS:
DESIGN AND IMPLEMENTATION OF AN INTELLIGENT AGRICULTURAL PRODUCE DISTRIBUTION SYSTEM

BY

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DECLARATION

We, Bissue Nana Kobina Ekoma, Arthur Lawrencia Ebela, Yirenkyi Bernice Agyeiwaa, Amissah Evans and Amprofi Emmanuel Wobir Kojo declare that this project work is our work. It is being submitted for the degree of Bachelor of Science in Computer Science and Engineering in the University of Mines and Technology (UMaT), Tarkwa. It has not been submitted for any degree or examination in any other University.

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DEDICATION

To the smallholder farmers of Ghana and Sub-Saharan Africa, whose labour feeds nations yet whose efforts are too often diminished by systems beyond their control. This work is a humble step toward a future where technology places the tools of fair trade directly in your hands.

To our families, whose encouragement and patience sustained us through every stage of this project.

ACKNOWLEDGEMENTS

We thank God for His guidance throughout this project. We express our sincerest gratitude to Dr Emmanuel Effah for his supervision, constructive advice and inputs throughout the development of AgroNexus, and to the University of Mines and Technology for providing an enabling learning environment. We also thank our colleagues, peers, families and friends for their encouragement, support and criticism during this process.

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LIST OF ABBREVIATIONS

AI	Artificial Intelligence
API	Application Programming Interface
B2B	Business-to-Business
B2C	Business-to-Consumer
CSS	Cascading Style Sheets
eNAM	Electronic National Agriculture Market
FAO	Food and Agriculture Organization of the United Nations
GPS	Global Positioning System
ICT	Information and Communications Technology
ILO	International Labour Organization
IoT	Internet of Things
JWT	JSON Web Token
MAPE	Mean Absolute Percentage Error
MIS	Market Information System
ML	Machine Learning
MOFA	Ministry of Food and Agriculture
PHL	Post-Harvest Loss
RBAC	Role-Based Access Control
REST	Representational State Transfer
RMSE	Root Mean Square Error
SMS	Short Message Service
SSA	Sub-Saharan Africa
UI	User Interface
UMaT	University of Mines and Technology
UX	User Experience
WFP	World Food Programme

CHAPTER 1

INTRODUCTION

1.1 Background

Across Ghana and much of Sub-Saharan Africa, farming forms the foundation on which most rural households depend for their survival. The agricultural sector accounts for well over sixty percent of employment across the region (ILO, 2024). In Ghana specifically, the Ministry of Food and Agriculture estimates that the sector contributes approximately twenty percent of gross domestic product and employs around fifty-two percent of the workforce, with smallholder farmers producing the majority of domestically consumed food (MOFA, 2025). Yet despite this central role, the sector haemorrhages value at nearly every stage between harvest and final consumption. Post-harvest losses in Sub-Saharan Africa range between twenty and fifty percent of key commodities depending on the crop and season, driven by inadequate storage infrastructure, poor rural road networks and the absence of reliable market information channels (Tanle *et al.*, 2025). The World Food Programme estimates that the annual financial cost of these losses across sub-Saharan Africa exceeds four billion US dollars, surpassing the total food aid the region receives in any given year (WFP, 2024). When one considers the labour, inputs and time behind every bag of tomatoes or crate of plantain, this figure represents a staggering proportion of rural income that simply vanishes.

The scale of the loss, however, conceals a more fundamental problem: the people and resources needed to prevent it are often present, but are unable to find one another. Farmers, buyers and transporters operate in isolation, each holding information the others need but with no shared channel through which to communicate, agree prices and move goods before they spoil (Amadu and McNamara, 2024; Wang *et al.*, 2025). The result is that a farmer in one district can sit on a surplus of tomatoes while a buyer in the neighbouring district faces a genuine shortage, not because the supply does not exist, but because no digital mechanism exists to connect the two.

Recent advances in artificial intelligence and mobile technology have opened a practical route to closing this coordination gap. Demand forecasting models built on sequential learning architectures can signal where commodity shortfalls are likely before they materialise, giving farmers a concrete basis for planting and harvest timing decisions (Elufioye *et al.*, 2024). Real-time price dashboards reduce the information asymmetry that allows middlemen to extract value from farmers who lack market visibility. Transport coordination tools can match farmers to available vehicles on the same day an order is placed (Das *et al.*, 2025; Ikade *et al.*, 2024). AgroNexus brings these capabilities together into a single platform built specifically for agricultural stakeholders in Ghana's Western Region.

1.2 Statement of Problem

Ghana's agricultural distribution system suffers from three interconnected failures, each of which can be addressed through digital coordination tools (Abate *et al.*, 2023; Izdori *et al.*, 2025).

Information failure. Farmers decide what to grow, when to bring produce to market, and where to sell without access to reliable demand data. Decisions are made on the basis of last season's prices and informal word of mouth. The result is that produce regularly arrives at markets already glutted, forcing distress sales below the cost of production (MOFA, 2025; Tanle *et al.*, 2025). This outcome is not recklessness on the part of farmers. It is the rational response to an information vacuum that a digital platform is specifically designed to fill.

Coordination failure. Even when a willing buyer exists, completing the transaction proves difficult. Farmers, traders, transporters and consumers operate without a common platform through which to communicate, negotiate and confirm delivery. The few digital tools available in Ghana tend to serve only one segment of this chain, typically buyers seeking price information, without integrating the full sequence from listing to payment confirmation (Izdori *et al.*, 2025). Amadu and McNamara (2024) found that structured coordination among supply chain actors consistently reduces losses, and that the absence of a shared coordination mechanism is the most significant reason why technology interventions in Ghanaian agriculture have fallen short of expectations.

Technology access failure. Solutions designed for high-speed internet connections and high-specification smartphones exclude the very farmers they were built to serve. Across the Western Region, the majority of smallholder farmers use entry-level Android handsets on patchy 2G and 3G networks (Das *et al.*, 2025). A platform that stops functioning at the farm gate is not a solution. AgroNexus is designed to operate under precisely these conditions.

1.3 Aim and Objectives

The primary aim of this project is to design and implement AgroNexus, an intelligent agricultural produce distribution system that optimises the coordination of supply and demand between farmers, consumers and transporters through AI-powered demand forecasting, real-time market price data and an accessible web and mobile interface.

The specific objectives of the project are to:

i. design and develop a multi-role web and mobile platform enabling farmers to list produce with pricing and availability details, consumers to browse and place orders, and transporters to receive and accept delivery assignments, with secure role-based access control and user management enforced across all platform functions;

ii. implement a real-time market intelligence dashboard providing current commodity price trends, regional supply levels and demand activity to support informed decision-making for all user roles;

iii. develop and train a Long Short-Term Memory (LSTM) neural network to generate rolling seven-day regional demand forecasts per crop type, targeting a Mean Absolute Percentage Error of twenty percent or below, corresponding to the eighty percent accuracy threshold stated in the aim; and

iv. create a transport coordination module that, upon receipt of a consumer order, automatically generates a delivery request visible to registered transporters in the relevant region, enabling timely collection and delivery of produce.

1.4 Tools and Facilities Used

The tools and facilities used in this project include:

i. React.js (web application) and Flutter (mobile application);

ii. Node.js and Express.js for server-side logic and API development;

iii. Supabase for relational data storage, authentication and real-time capabilities;

iv. Python, TensorFlow and Scikit-learn for demand forecasting model development;

v. Git, Docker, Jest and PyTest for version control, containerisation and testing;

vi. Internet access and library resources; and

vii. Project supervision from the Department of Computer Science and Engineering, UMaT.

1.5 Methods Used

The methods used in this project include:

i. review of relevant literature on agricultural distribution, post-harvest loss and AI-driven supply chain systems;

ii. requirements gathering from the perspectives of farmers, traders and transporters;

iii. system and project design following an Agile-Iterative Software Development Methodology combined with User-Centred Design principles;

iv. AI model development, training and iterative validation;

v. integration and system testing across functional, performance and usability dimensions; and

vi. deployment and final documentation.

1.6 Scope of Work

AgroNexus is scoped to address the software and intelligence layers of agricultural distribution. The system covers user registration and role-based access for farmers, consumers and transporters; supply and demand management modules; AI-powered demand forecasting and allocation optimisation; a transport coordination module; a real-time dashboard with price information; and a mobile application with offline functionality.

Payment processing is deliberately excluded from the scope of this project. Offering payment services in Ghana requires licensing from the Bank of Ghana under the Payment Systems and Services Act, 2019 (Act 987), and compliance with Payment Card Industry Data Security Standards (PCI-DSS). Meeting these obligations introduces regulatory, legal and security architecture requirements that lie outside the scope of a software coordination system. AgroNexus is developed to demonstrate that the harder coordination problem, connecting farmers, consumers and transporters in real time and using machine learning to anticipate demand patterns, is solvable. Payments represent the logical next step once the coordination and distribution functions are verified to be working reliably.

Physical storage management and IoT hardware deployment are similarly beyond the current scope. AgroNexus functions as a software and intelligence layer rather than a logistics infrastructure project. The prototype is validated in a controlled environment, with the architecture designed to support full deployment once validation is complete.

1.7 Expected Outcomes

i. *A Unified Distribution Platform:* A web and mobile application through which farmers list produce, consumers place orders and transporters accept delivery requests within a single coordinated system.

ii. *Improved Market Intelligence:* A real-time dashboard displaying commodity prices and supply and demand data across regions, reducing information asymmetry for all stakeholders.

iii. *AI-Driven Demand Forecasting:* An LSTM-based machine learning module generating rolling seven-day regional demand forecasts per crop type, targeting a MAPE of twenty percent or below, enabling proactive distribution decisions that reduce post-harvest losses.

iv. *Transport Coordination:* A module linking farmers directly to available transporters in the relevant region, reducing the logistical delays that contribute to produce wastage.

1.8 Work Organisation

This report covers the first three chapters of the project. Chapter 1 provides the background, problem statement and objectives. Chapter 2 presents the literature review covering post-harvest loss, existing digital agricultural platforms, AI in supply chains and mobile application design for low-resource contexts. Chapter 3 describes the methodology adopted for the project, followed by the system design, including architecture, AI model design and data flow.

CHAPTER 2

LITERATURE REVIEW

The literature reviewed in this chapter is organised around five themes that together frame the problem AgroNexus is designed to address. The chapter begins with the scale and root causes of post-harvest loss in developing agricultural economies, then examines the record of digital agricultural platforms that have attempted to solve related problems. It then reviews the evidence on machine learning for agricultural demand forecasting, considers the broader literature on e-commerce and multi-stakeholder coordination in agriculture, and concludes with research on mobile application design for low-resource settings. Each section draws out specific lessons that shaped the design of AgroNexus.

2.1 Post-Harvest Losses in Developing Agricultural Economies

Post-harvest loss is widely recognised as one of the most significant obstacles to food security and farmer income in Sub-Saharan Africa (Stathers *et al.*, 2024; WFP, 2024). This section examines the scale of the problem, the root causes that storage-focused interventions have failed to address, and the role of information asymmetry in sustaining the losses that persist across agricultural supply chains in Ghana and the wider region.

2.1.1 The Scale of the Problem

Post-harvest loss refers to any reduction in the quantity or quality of food between harvest and the point of consumption. On a global scale, roughly one-third of all food produced for human consumption is either lost or wasted each year, amounting to approximately 1.3 billion tonnes (FAO, 2019). Sub-Saharan Africa sits above this global average. Depending on the particular crop and the season in question, losses in the region range between twenty and fifty percent of key commodities (Tanle *et al.*, 2025). The World Food Programme estimates that the annual financial cost of food losses across Sub-Saharan Africa exceeds four billion US dollars, a sum that surpasses the total food aid the region typically receives in any given year (WFP, 2024).

In Ghana specifically, post-harvest losses of vegetables and other perishable crops are among the most severe, driven by the combination of inadequate cold chain infrastructure, unreliable electricity supply in rural markets and inconsistent post-harvest handling practices (Essilfie and Baddoo, 2025). These figures carry very real human weight. A smallholder farmer who loses a third of her tomato harvest does not simply lose a third of her profit. Once transport costs, hired labour and inputs have been subtracted, she may well lose all of it. And when supply drops, consumer prices rise, pushing food further out of reach for households least able to absorb such increases. Smallholder farmers in Ghana, who produce most of the food consumed domestically, face this pressure from both directions: each loss at harvest reduces their income while tighter supply elsewhere raises the cost of what they themselves must buy (MOFA, 2025).

2.1.2 Root Causes Beyond Storage

The standard response to post-harvest loss has traditionally been better storage. Cold chain facilities, hermetic bags and improved grain silos dominate the landscape of post-harvest loss interventions, yet the evidence shows that this approach addresses only one dimension of a multi-layered problem (Stathers *et al.*, 2024). Storage technology solves the problem of deterioration in the hours after harvest. It does nothing to solve the problem of a farmer who has no confirmed buyer, no available vehicle and no visibility into which market is currently short of her particular crop.

Research into South African vegetable supply chains found that twenty-five percent of farmers identified lack of transport as the direct cause of their losses, while a further twenty-six percent cited distance from market (Qange *et al.*, 2024). A better storage bag solves neither of these issues. Produce rots at the farm gate because no vehicle ever arrives. Crops reach market without any buyer confirmed in advance, forcing farmers to accept whatever price happens to be on offer that day, often below the actual cost of production. Tanle *et al.* (2025) confirm that this pattern of logistics-driven loss is consistent across Sub-Saharan Africa and is the aspect of the post-harvest problem least addressed by existing interventions.

There is also a spatial dimension to consider. Across Ghana and the broader region, surplus and shortage frequently coexist in neighbouring districts, with nothing in place to move supply from one area to another efficiently. Wang *et al.* (2025) describe this mismatch as one of the most persistent structural weaknesses in developing agricultural markets. Stathers *et al.* (2024) note that coordination among farmers, traders, transporters and buyers remains the area most consistently underinvested in both research and development programmes.

Figure 2.1 classifies these root causes into the three interlocking categories that the literature consistently identifies (Tanle *et al.*, 2025; Qange *et al.*, 2024; Wang *et al.*, 2025). Each branch represents a distinct failure mode; together they explain why improving storage alone has never been enough to close the post-harvest loss gap in Sub-Saharan Africa.

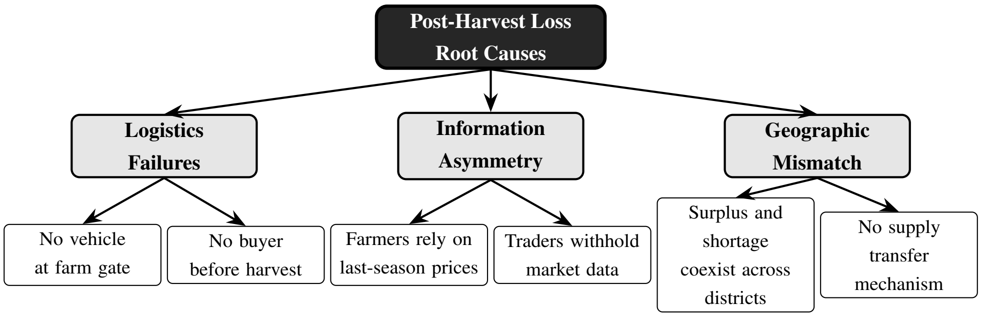

Figure 2.1: Classification of Root Causes of Post-Harvest Loss in Developing Agricultural Economies (Source: Author's Analysis based on Tanle *et al.*, 2025; Qange *et al.*, 2024; and Wang *et al.*, 2025)

2.1.3 The Role of Information Asymmetry

Beneath both the logistics problem and the geographic mismatch lies a more fundamental issue: no single actor in the chain can see what the others actually know. A farmer decides what to plant and when to harvest based on prices from the previous season and whatever local word of mouth reaches her. A trader knows which markets are active but has no structured way to share this knowledge with the farmers supplying her. A transporter waits without knowing that a load is ready nearby (Amadu and McNamara, 2024). Each actor makes a decision that seems locally rational, but when combined with the equivalent decisions of everyone else in the chain, the collective outcome is waste.

This dynamic has been documented specifically in Ghanaian agricultural markets. Sarku and Ayamga (2025) found that digital platform operators in Ghana have often benefited more from the data they collect from farmers than the farmers themselves, precisely because the information asymmetry between platform operators and smallholder users is never resolved. Amadu and McNamara (2024) demonstrated that structured information-sharing among supply chain actors consistently produces better outcomes than each party operating independently. Izdori *et al.* (2025) went further, concluding that the absence of shared coordination mechanisms represents the single most important reason why technology interventions in agriculture have so often fallen short of expectations. For AgroNexus, the implication is clear: the platform must place information that each actor currently holds in isolation into the hands of all three groups simultaneously.

2.2 Digital Agricultural Platforms

A number of platforms have attempted to connect farmers with markets across Africa, Asia and beyond. This section examines several case studies relevant to the AgroNexus design: the general landscape of AgriTech platforms and why most have struggled to scale; the Ghanaian market information system Esoko; Twiga Foods in Kenya; the mFarm marketplace; KhetBazaar in India; and the national-scale eNAM and AgriBazaar platforms, also in India. Each case contributed a specific design lesson to AgroNexus.

2.2.1 The Promise and Limitations of AgriTech Platforms

Mobile platforms linking farmers to markets have existed for well over a decade across Africa, India and Southeast Asia, yet the track record is decidedly mixed. A 2024 review of agricultural marketplace platforms across developing countries found that nearly all pure-play digital matchmakers had been forced to overhaul their business models within a few years of launch, facing the same obstacles regardless of setting: agricultural revenue arrives seasonally rather than steadily; individual transactions are small; supply chains remain fragmented; and rural infrastructure proves unreliable at precisely the worst moments (ISF Advisors, 2024).

In Ghana specifically, Sarku and Ayamga (2025) found that while numerous digital platforms have been developed for the smallholder farming sector, ranging from supply chain management tools to livestock and aquaculture platforms, most have served platform operators better than the farmers they were built for. The study documents platforms such as AgroCenta's CropChain for supply chain management, DigiExt for buyer-matching, and TrotroTractor for equipment access, confirming that individual platform categories exist in Ghana but that no single platform integrates the full chain. Those platforms that survived the initial years of operation did not remain pure matchmakers. They integrated into logistics, provided financing, aggregated supply or became buyers themselves (ISF Advisors, 2024). The lesson for AgroNexus is straightforward: a farmer in the Western Region already has informal ways to locate a buyer. What she lacks is a single place to find that buyer, agree a price, arrange the vehicle and track the delivery as part of one continuous interaction.

2.2.2 Esoko (Ghana)

Esoko has operated in Ghana since 2005, longer than almost any comparable agricultural information service on the continent. It delivers commodity prices, weather forecasts and farming advice via SMS and voice call in local languages, deliberately targeting farmers with basic handsets rather than smartphones (Engineering for Change, 2024). Within its particular scope, Esoko works. Farmers receiving price alerts sold their produce at roughly ten percent higher prices than those without access, and an evaluation found a nine percent rise in income along with an eighteen percent improvement in profit margins among users (Abate *et al.*, 2023).

Yet what Esoko cannot do defines precisely the gap that AgroNexus seeks to address. A farmer who learns that tomatoes are fetching a good price in Kumasi must still find the buyer in Kumasi, agree terms, load the vehicle and confirm collection, none of which Esoko facilitates (Engineering for Change, 2024). The price becomes visible but the transaction remains just as difficult as before. This was a deliberate design choice suited to what SMS technology can deliver. The coordination problem of finding both the buyer and the truck at the same time, within the window before perishable produce deteriorates, remains unresolved.

2.2.3 Twiga Foods (Kenya)

Twiga Foods attempted the opposite approach. Founded in Nairobi in 2014, the company went well beyond simply matching farmers with buyers. It sourced produce directly, ran its own quality inspections, operated distribution centres and maintained a delivery fleet. At its peak in 2022, roughly two million kilograms of fresh produce moved through its network daily across twelve cities, reaching some one hundred and forty thousand retail vendors (Finnfund, 2024). Post-harvest losses for the crops it handled dropped from around thirty percent to just four percent.

Then the fixed costs caught up. In 2023, Twiga shed approximately two hundred and eighty-three employees, shuttered several warehouses and outsourced delivery to third-party operators (Afrobility, 2025). Owning trucks, warehouse space and quality control staff carries a cost base that agricultural margins cannot sustain through a poor season. ISF Advisors (2024) cited Twiga as an illustration of how capital-intensive integration fails financially even when the underlying transactions are economically sound. The Twiga experience settled a key design question for AgroNexus: the platform does not own logistics infrastructure. Independent transporters in the Western Region already exist and already move agricultural goods; AgroNexus makes it easier for a farmer to find one the moment an order comes in, without absorbing the liability of running a fleet.

2.2.4 mFarm and Similar Mobile Marketplaces

The mFarm platform in Kenya gave farmers a way to list their crops and connect directly with buyers, bypassing the chain of middlemen that typically absorbs a large share of farm gate revenue. Studies of direct-to-buyer digital platforms found that farmers who bypassed traditional middlemen reported income increases of between twenty and thirty percent (KIPPRA, 2023). That represents a significant gain simply from removing intermediary margins.

The limitation proved consistently the same regardless of geography. Even when a farmer had a listing and a buyer expressed interest, the absence of any integrated transport function meant that perishable goods frequently spoiled before the transaction could actually be completed (KIPPRA, 2023). Muromba *et al.* (2025) confirm that this gap between market matching and physical delivery coordination is the most frequently cited shortcoming of agricultural marketplace applications across Sub-Saharan Africa. A buyer who sees tomatoes listed at an attractive price cannot always arrange collection within the narrow window before the produce deteriorates. What is needed is not merely a marketplace but a marketplace with a logistics layer built in.

2.2.5 KhetBazaar (India)

KhetBazaar is a mobile agricultural marketplace developed in India that organises its functions into three separate modules: a farmer module for listing and managing produce, a buyer module for browsing and placing orders, and an admin module for platform management. This role-specific design means that each user type sees only the features relevant to them, simplifying the interface and reducing confusion during onboarding (Maatman and Rosenstock, 2024). The platform processes transactions in real time and includes a multilingual interface that has improved uptake among farmers with limited English literacy.

KhetBazaar demonstrates that role-based interface design works well in agricultural contexts and that multilingual support serves as a meaningful adoption driver (Maatman and Rosenstock, 2024). Both principles find their way into the AgroNexus design. However, KhetBazaar does not include demand forecasting and does not coordinate transport. A farmer using KhetBazaar knows what crops are currently in demand but has no forecast of what demand will look like next week and no way to arrange delivery through the same platform. Both functions sit at the heart of what AgroNexus provides.

2.2.6 eNAM and AgriBazaar (India)

India's Electronic National Agriculture Market (eNAM) and the AgriBazaar platform have processed agricultural trade at national scale and include tools for price discovery, digital payment and some degree of demand data. Their breadth is genuinely impressive and represents one of the most ambitious attempts anywhere in the world to digitise agricultural markets (Abate *et al.*, 2023).

Yet studies of both platforms found that adoption remained limited among the very smallholder farmers they were most intended to help. The main barriers cited were interface complexity, poor performance on low-specification devices and the absence of logistics integration (Abate *et al.*, 2023). A farmer who needs multiple steps and a reliable smartphone data connection to complete a transaction will default to her existing middleman, however unfavourable the terms. A platform can possess every feature farmers need and still see low adoption if it proves too slow or too complicated to use on the devices farmers actually own.

2.3 Machine Learning in Agricultural Supply Chain Forecasting

Demand forecasting is central to the AgroNexus value proposition, and machine learning has emerged as the most effective approach for this task in agricultural settings. This section reviews the growth of the field, the suitability of time series models for agricultural demand data, the specific case for Long Short-Term Memory

(LSTM) networks, and the relationship between forecasting accuracy and supply chain performance.

2.3.1 Growth of the Field

Research on machine learning applications in agriculture and agricultural supply chains has expanded rapidly since 2020. A bibliometric review of papers on the topic spanning 2015 to 2024 showed that approximately three-quarters of everything published in this area appeared after 2021 alone (Mele *et al.*, 2024). This rapid growth reflects both the increasing availability of agricultural data and the falling computational cost of training deep learning models. A separate survey counted over twenty distinct machine learning approaches applied across crop management, disease detection, price forecasting and distribution problems in agriculture (Elufioye *et al.*, 2024).

2.3.2 Why Time Series Models Suit Agricultural Demand Data

Agricultural demand data possesses a structure that distinguishes it from most other forecasting problems. Demand for crops is not random. It follows seasonal cycles tied to planting seasons, harvest periods and calendar events such as festivals and school terms. It responds to weather, which itself follows predictable seasonal patterns. It reacts to price movements, and prices in turn respond to supply levels, creating feedback loops that repeat across years. Because the data is ordered over time and contains repeating patterns at multiple scales, it is well suited to time series forecasting methods specifically designed to learn from sequential data (Elufioye *et al.*, 2024).

Manogna *et al.* (2025), using daily wholesale price data for twenty-three agricultural commodities, found that deep learning models treating price and demand data as ordered sequences consistently outperformed traditional stochastic models across all error metrics. Their results confirmed that LSTM networks are particularly effective for highly volatile commodity data because of their ability to retain information across long time spans. Earlier work by Kantasa-Ard *et al.* (2021) applied time series machine learning to agricultural products in Thailand and demonstrated that forecasts produced this way helped distribution networks reduce waste by matching supply more closely to anticipated demand. Research in Morocco applied time series machine learning to historical agricultural sales records, incorporating seasonal indicators, climate data and location variables, and found that the resulting model was accurate enough to meaningfully improve supply planning at a regional level (Nebri *et al.*, 2024). Studies comparing multiple forecasting methods across major crops found that time series models using both supply volume and market price as inputs outperformed approaches that used either variable in isolation (Elufioye *et al.*, 2024).

These findings support the decision to build the AgroNexus demand forecasting module as a time series model. The data the platform collects, including daily order volumes, transaction prices, crop types and region identifiers, is exactly the kind of sequential, seasonal, multi-variable data that time series approaches handle well.

2.3.3 LSTM Networks for Agricultural Forecasting

Among the available time series modelling approaches, Long Short-Term Memory (LSTM) networks have emerged as a strong choice for agricultural forecasting tasks. An LSTM is a type of recurrent neural network that processes sequences step by step while maintaining a form of memory across steps. This allows the model to capture not just the most recent trend but also longer patterns that span weeks or months. For agricultural demand, this matters because a crop such as maize or groundnut may have both a short-term price trend and a longer seasonal cycle, and a good forecast needs to account for both (Dhal and Kar, 2024).

Manogna *et al.* (2025) confirmed empirically that LSTM networks outperform traditional statistical models including ARIMA and Support Vector Regression when applied to agricultural commodity price data, particularly for highly volatile crops where seasonal and weather-driven patterns are pronounced. The superiority of LSTM over simpler models in this context has been attributed to its ability to selectively retain or discard information across time steps through its gating mechanism, which prevents the model from being overwhelmed by short-term noise when longer-term seasonal patterns are the dominant driver of demand. Elufioye *et al.* (2024) found that hybrid approaches combining LSTM networks with gradient boosting models for tabular feature processing performed particularly well in agricultural supply chain contexts. This two-stage approach is the model architecture adopted for AgroNexus.

2.3.4 Machine Learning for Supply Chain Optimisation

Beyond demand forecasting, machine learning has been applied to the broader question of how to allocate crops across a distribution network once demand is known. Dhal and Kar (2024), reviewing AI applications in food security and supply chain management, found that systems deployed in data-poor environments benefit from a staged approach to model deployment, starting with simpler and more interpretable models and moving toward more complex architectures as additional data becomes available.

El Bhilat *et al.* (2024) confirmed empirically that in agri-food supply chains, the accuracy and impact of machine learning predictions depends significantly on the efficiency of the distribution network they feed into. A good forecast paired with an unreliable logistics layer produces worse outcomes than a mediocre forecast paired with a well-organised logistics layer. This finding reinforces the core design decision in AgroNexus to develop the demand forecasting module and the transport coordination module as complementary parts of the same system rather than as independent features.

2.4 E-Commerce and Multi-Stakeholder Coordination in Agriculture

Beyond individual platform case studies, the broader literature on agricultural e-commerce and multi-stakeholder coordination offers evidence about what platform features drive adoption and what structural conditions produce the best outcomes for farmers. This section reviews that evidence and draws the implications for the design of AgroNexus.

2.4.1 E-Commerce Platforms in Sub-Saharan Africa

A systematic review covering agricultural e-commerce platforms in Sub-Saharan Africa across the period 2010 to 2024 found that digital marketplaces do improve market access and lead to better prices and lower post-harvest losses where they are successfully adopted (Mele *et al.*, 2024). The review found that no single platform had successfully combined market matching, logistics coordination and demand intelligence for smallholder farmers in the region. The barriers to adoption that kept emerging across all categories were limited digital literacy, inconsistent internet connectivity and the absence of trusted intermediaries to help farmers make the transition to digital selling.

In Ghana, these barriers have been directly observed. Sarku and Ayamga (2025) found that even among the growing number of digital platforms operating in Ghana's agricultural sector, adoption by smallholder farmers remains constrained by interface complexity, data costs and a lack of offline functionality. A CGAP review of agricultural marketplace platforms in Africa found that platforms combining market matching with logistics coordination outperform those handling only one function, and that logistics costs represent the biggest operational challenge for agricultural platforms in Africa (CGAP, 2023). Without a working logistics layer, even a well-designed marketplace cannot reliably deliver on its promise to farmers.

Abate *et al.* (2023), studying the trajectory of digital tools for agricultural market transformation in Africa, identified three things that farmers consistently said they needed from a digital platform: a reliable way to find buyers, a way to arrange delivery, and some indication of what demand would look like in the near future. They found that most available platforms addressed one or at most two of these three needs. AgroNexus is designed to address all three within a single platform.

2.4.2 Multi-Stakeholder Coordination

Izdori *et al.* (2025) developed a formal framework for multi-stakeholder coordination in agricultural supply chains focused specifically on reducing post-harvest losses. Their work demonstrated that losses decline when actors at different levels of the supply chain share information and coordinate decisions through a shared platform. The study identified three mechanisms through which a shared platform creates value: it gives each actor visibility into what others are doing; it creates accountability because transactions are recorded; and it enables forward planning because all actors can see aggregate demand and supply data rather than just their own slice of it.

Amadu and McNamara (2024) tested a related hypothesis, finding that structured coordination among agricultural stakeholders produced better post-harvest outcomes and that the gains were largest for farmers in the middle of the supply chain who possessed the least individual power to influence prices or logistics. Wang *et al.* (2025) corroborate this finding, noting that digital coordination tools create the most value not for the largest and most resourced actors but for those who have historically been excluded from market information. This result supports the case for designing AgroNexus as a platform that gives farmers access to information and coordination tools that currently only traders and large buyers possess.

2.5 Mobile Application Design for Low-Resource Settings

The design choices made in the AgroNexus mobile application are grounded in research on what works for smallholder farmers in low-resource environments. This section reviews the constraints of the target user base, the rationale for selecting Flutter as the development framework, and the evidence on what drives sustained adoption among agricultural communities.

2.5.1 The User Base and Its Constraints

Designing a mobile application for a smallholder farmer in Bogoso or Prestea is a fundamentally different exercise from designing one for a professional in Accra. The target users of the AgroNexus mobile application work outdoors, frequently in areas with patchy data coverage, and are likely using entry-level Android handsets that are two to five years old (Das *et al.*, 2025). Muromba *et al.* (2025) found that across Sub-Saharan Africa, the most consistently cited barriers to mobile agricultural application adoption are interface complexity, slow load times on low-specification devices and the absence of offline functionality, with these three factors accounting for the majority of abandonment cases across the platforms reviewed.

The research on this point is clear. Applications built for higher-specification devices with layered menus consistently fail to gain traction among smallholder farmers, regardless of how useful the content inside them might be (Das *et al.*, 2025). In Kenya, where mobile internet is widely available, only twenty to thirty percent of farmers adopted digital agricultural tools; the barriers were slow load times and interfaces too complex to navigate quickly (KIPPRA, 2023). An application that stops working at the farm gate, where connectivity drops, is not genuinely useful to the people it was built for.

2.5.2 Why Flutter Is the Right Choice

Flutter was chosen as the mobile framework because of the specific constraints described above. Unlike frameworks that execute through a JavaScript bridge, Flutter compiles to native machine code for both Android and iOS. On a two-year-old entry-level Android handset, the difference in execution speed is noticeable in daily use and directly affects whether a user continues using an application or abandons it (Muromba *et al.*, 2025). A single codebase producing both platform builds keeps development effort within reach for a student team while ensuring that features remain consistent across Android and iOS.

Flutter's widget system makes it practical to build the kind of interface a farmer at Tarkwa market actually needs: large touch targets, text readable at a glance, and navigation shallow enough to reach any core action in just a few taps. These are not cosmetic preferences. They are what separates a tool that gets picked up from one that gets abandoned, particularly for users with limited prior experience of smartphone applications (Muromba *et al.*, 2025; Das *et al.*, 2025).

For connectivity, Flutter's local storage libraries allow recent data to be cached on-device so that farmers can browse listings, check forecasts and view order status without an active connection. Writes made while offline are queued and pushed to the server the moment connectivity returns, requiring no action from the user. Muromba *et al.* (2025) identified offline functionality as one of the most critical factors in sustained adoption of mobile agricultural applications in Sub-Saharan Africa, making this a design requirement rather than an optional enhancement.

2.5.3 Trust, Integration and Reliability as Adoption Drivers

Wang *et al.* (2025) studied what separates agricultural platforms that sustain adoption from those that get abandoned. Three factors consistently made the difference. Farmers had to trust that information was accurate and that transactions would complete as agreed. The platform had to draw from the same data sources that other chain actors used, rather than existing as an isolated island. And it had to be stable enough that farmers could count on it at the moments that mattered.

Each of these three requirements maps directly to a design choice in AgroNexus. Trust is built through role-based access control, where each user can only see and act on what their role permits, and through order tracking that is visible to all parties throughout (Izdori *et al.*, 2025). Integration is achieved by running everything through a single Supabase database: the dashboard, the forecasting model and the transport module all read from the same source. Reliability rests on the offline-first mobile design, which keeps the application usable when data coverage drops, and on Supabase's real-time synchronisation, which keeps the web and mobile clients consistent without manual reconciliation.

2.6 Review of Related Works

Table 2.1 below summarises the key works reviewed in this chapter, showing the methodology used in each, the main contribution it made and the specific limitation that AgroNexus is designed to address.

Table 2.1: Review of Related Works

Author / System	**Methodology**	**Key Contribution**	**Limitation Addressed by AgroNexus**
Esoko, Ghana (Engineering for Change, 2024)	SMS and voice market information system	Farmers using price alerts sold produce at 10% higher prices with reach across 20 African nations	No transaction facilitation, transport coordination or demand forecasting
Twiga Foods, Kenya (Finnfund, 2024)	B2B mobile platform with owned delivery fleet	Post-harvest losses reduced from 30% to 4% across 140,000 retail vendors	Owning logistics proved too costly; platform excluded direct consumer access
mFarm, Kenya (KIPPRA, 2023)	Mobile farmer-to-consumer marketplace	Direct selling to buyers raised farmer income by up to 30%	No demand forecasting or transport coordination; perishables spoil before delivery
KhetBazaar, India (Maatman and Rosenstock, 2024)	Role-specific mobile marketplace modules	Real-time farmer-buyer connectivity with a multilingual interface	No demand forecasting, no transport coordination and no regional supply dashboard

Author / System	**Methodology**	**Key Contribution**	**Limitation Addressed by AgroNexus**
eNAM / AgriBazaar, India (Abate *et al.*, 2023)	National-scale digital trading platform	Large-scale agricultural trade with price discovery tools	Interface complexity limits smallholder adoption; logistics remain unintegrated
Sarku and Ayamga (2025)	Analysis of digital platform practices in Ghana	Documented the landscape of active digital platforms in Ghana's smallholder farming sector	Confirms no single Ghanaian platform integrates listing, ordering, transport and forecasting
Elufioye *et al.* (2024)	Review of ML in agricultural supply chains	Confirmed time series ML value in demand forecasting and logistics optimisation	Theoretical review only; no implemented system
Manogna *et al.* (2025)	Empirical comparison of ML and DL models for agricultural price forecasting	Demonstrated LSTM superiority over traditional models for volatile agricultural commodity data	Focused on price forecasting only; no marketplace or transport integration
Dhal and Kar (2024)	Review of AI in food security and supply chains	Staged model deployment recommended for data-poor settings	No marketplace or transport integration provided
Stathers *et al.* (2024)	Systematic review of PHL interventions in SSA	Storage technology dominates research while logistics losses remain unaddressed	No digital platform or distribution coordination component

Author / System	Methodology	Key Contribution	Limitation Addressed by AgroNexus
Izdori *et al.* (2025)	Multi-stakeholder PHL reduction framework	Structured coordination among supply chain actors measurably reduces losses	Framework only; no platform implementation or machine learning component

Looking across the table, a clear pattern emerges. Platforms that inform do not transact. Platforms that transact do not move goods. Platforms that move goods do not forecast. Frameworks that diagnose the coordination problem do not build the system to solve it. Each existing solution addresses one part of what the problem actually demands. AgroNexus brings all of these parts together into a single platform, built around the specific conditions of smallholder farming in Ghana, including the Western Region markets, the connectivity constraints and the commodity mix that characterise this context.

2.7 Open Issues and Remedies

The works reviewed in this chapter, taken together, reveal a set of recurring gaps that no existing platform or study has fully addressed. This section names those gaps directly and explains how AgroNexus is designed to close each one.

2.7.1 Fragmentation Across the Supply Chain

Every platform reviewed operates on just one segment of the agricultural supply chain. Esoko informs but does not transact. mFarm and KhetBazaar facilitate buying and selling but leave transport to chance. Twiga handled the full chain but at a cost structure that proved unsustainable. No existing solution brings farmers, consumers and transporters into a single coordinated system, and the consequence is that even when individual links in the chain improve, the chain as a whole continues to break down at the handover points (Izdori *et al.*, 2025). Sarku and Ayamga (2025) confirm that this fragmentation characterises the Ghanaian platform landscape specifically, where numerous single-function platforms coexist without any integration between them.

AgroNexus addresses this fragmentation by integrating all three user groups into one platform. A consumer order automatically triggers a transport request visible to registered transporters in the relevant area. Notifications flow to all parties at each stage. No handover requires a separate communication channel outside the system.

2.7.2 Absence of Demand Forecasting

None of the platforms reviewed, whether Esoko, mFarm, Twiga, KhetBazaar or eNAM, provide farmers with a forward-looking signal about what demand will look like in the coming days or weeks. Farmers make planting and harvest timing decisions in an information vacuum, and the result is the cyclical surplus-and-shortage pattern that drives post-harvest loss across the region (Tanle *et al.*, 2025).

The AI demand forecasting module in AgroNexus directly addresses this gap. Trained on WFP price records, MOFA production data and weather variables, and structured around Ghana's crop harvest calendar and festival patterns, the LSTM model generates a rolling seven-day demand forecast per crop type per region. This gives farmers a concrete basis for deciding when to bring produce to market and which markets to target, information that currently does not reach them until after the decision has already been made.

2.7.3 Technology Access and Device Constraints

The platforms that have achieved the most, namely eNAM and AgriBazaar, require reliable internet connections and capable smartphones that lie beyond the reach of smallholder farmers in rural Ghana. The result is a consistent adoption gap between the farmers these platforms were intended to serve and the farmers who actually use them. Muromba *et al.* (2025) found that adoption rates of mobile agricultural applications in Sub-Saharan Africa remain low precisely because most applications are built for hardware and connectivity conditions that do not reflect the realities of rural farming environments.

AgroNexus responds to this challenge with a Flutter mobile application compiled to native machine code, optimised for entry-level Android handsets and designed around the interface simplicity that characterises tools which actually get adopted in low-resource settings (Das *et al.*, 2025; Muromba *et al.*, 2025). Offline functionality ensures the application remains useful at the farm gate and at market, where connectivity is most likely to be unreliable.

2.7.4 Unsustainable Logistics Models

Twiga's experience demonstrated that owning the logistics layer produces exceptional results in loss reduction but creates a cost structure that agricultural margins cannot support at scale (Afrobility, 2025). Most other platforms avoided this problem by simply not coordinating transport at all, leaving farmers to arrange delivery independently after a sale is agreed.

AgroNexus sidesteps this tension by acting as a coordinator rather than an owner. Independent transporters register on the platform and receive visibility of open delivery requests in their operating area. The platform earns value by making the matching faster and more reliable, without taking on vehicle costs, warehouse leases or driver salaries. This structure keeps the logistics function operational while avoiding the financial fragility that ended Twiga's integrated model.

The contrast between Twiga's ownership approach and the AgroNexus coordinator model can be expressed in straightforward cost terms. When a platform owns its logistics infrastructure, it carries a fixed cost burden F that does not shrink when seasonal transaction volumes fall. Total platform cost takes the form shown in Equation (2.1):

$$C_{\text{owner}}(q) = F + V(q) \tag{2.1}$$

where q is transaction volume and $V(q)$ represents variable costs that rise with activity. This is precisely what caught up with Twiga: the fixed cost floor was too high to survive the revenue variability inherent in agricultural markets (ISF Advisors, 2024).

A coordinator model sidesteps this entirely. Because vehicles and storage are owned by independent operators, the platform's own cost structure reduces to a small per-transaction fee c plus negligible overhead ϵ, as shown in Equation (2.2):

$$C_{\text{coord}}(q) = c \cdot q + \epsilon \tag{2.2}$$

The break-even transaction volume q^*, below which ownership always loses money, is given by Equation (2.3):

$$q^* = \frac{F}{p - c} \tag{2.3}$$

where p is revenue per transaction. In practice, Ghana's agricultural markets spend a significant part of each year below this threshold. AgroNexus is built around the coordinator cost structure, which means the platform remains financially viable even when seasonal volumes are at their lowest.

2.7.5 Theoretical Frameworks Without Implementation

The academic literature on multi-stakeholder coordination and post-harvest loss reduction, including Izdori *et al.* (2025) and Stathers *et al.* (2024), provides well-developed frameworks for understanding why losses persist and what conditions would reduce them. What the literature does not provide is a working system that puts those frameworks into practice in a West African agricultural context.

AgroNexus provides the implementation that the literature points toward but stops short of delivering. The platform architecture directly reflects the coordination mechanisms Izdori *et al.* (2025) identify, including shared visibility, recorded transactions and aggregate demand data accessible to all actors, applied to the specific crops, markets and infrastructure conditions of Ghana's Western Region.

Figure 2.2 illustrates the multi-stakeholder coordination framework adapted from Izdori *et al.* (2025) that underpins the AgroNexus design.

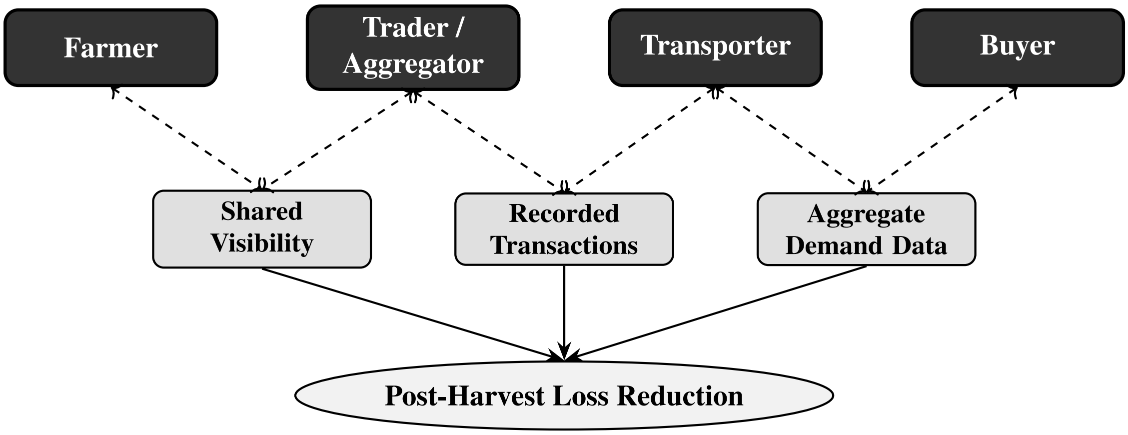

Figure 2.2: Multi-Stakeholder Coordination Framework for Post-Harvest Loss Reduction (Adapted from Izdori *et al.*, 2025)

CHAPTER 3
METHODOLOGY AND SYSTEM DESIGN

This chapter presents the methodology adopted for the development of AgroNexus and then describes the system design that results from applying that methodology. It begins with the software development approach and design philosophy, proceeds to the functional and non-functional requirements derived from the problem analysis, and then works through the architecture, the data pipeline for the demand forecasting model, the AI model design, the interface design choices, the technology stack, and the flow of data from a user action through to a visible result on screen.

3.1 Methodology Adopted

The development of AgroNexus follows an Agile-Iterative Software Development Methodology combined with User-Centred Design (UCD) principles. This section describes each component of the methodology and explains why the combination suits the particular challenges of building a multi-stakeholder agricultural distribution platform under the constraints identified in Chapters 1 and 2.

3.1.1 Agile-Iterative Software Development

The Agile-Iterative approach organises development into short, repeatable cycles called iterations, each of which produces a working increment of the system that can be tested, evaluated and refined before the next iteration begins (Tetteh, 2024). Unlike a linear waterfall model, where each phase must be completed in full before the next phase starts, the iterative model allows the team to revisit requirements, architecture and design decisions as understanding of the problem deepens through building and testing (Alsaqqa *et al.*, 2024). This is particularly important for AgroNexus because the platform serves three distinct user groups, namely farmers, consumers and transporters, whose needs interact in ways that only become fully apparent when working software is placed in front of representative users.

The choice of Agile-Iterative development is supported by three considerations drawn from the problem context. First, the coordination failures identified in Chapter 1 involve multiple interdependent user roles, and the interactions between roles cannot be fully specified in advance. An iterative approach allows the team to build one role's workflow, test it, and then refine the design of the next role's workflow in light of what was learned. Second, the literature reviewed in Chapter 2 showed that agricultural platforms which launched with a fixed feature set and no mechanism for iterative refinement consistently failed to achieve sustained adoption (ISF Advisors, 2024; Sarku and Ayamga, 2025). Third, the project team consists of five undergraduate developers working within a fixed academic timeline. Agile iterations with clearly scoped deliverables provide the structure needed to manage parallel workstreams across the frontend, backend and AI components without one stream blocking the others.

Each iteration in the AgroNexus development cycle follows a four-stage sequence: plan the iteration scope, implement the planned features, test the increment against acceptance criteria, and review the results to identify adjustments for the next iteration (Alsaqqa *et al.*, 2024). Iterations are time-boxed to two weeks, which aligns with the academic supervision schedule and ensures that progress is demonstrable at each supervisory meeting.

3.1.2 User-Centred Design

User-Centred Design is a design philosophy that places the needs, capabilities and limitations of end users at the centre of every design decision (ISO, 2019). For AgroNexus, this is not an optional enhancement but a structural requirement. The literature reviewed in Chapter 2 established that the primary reason agricultural platforms fail to achieve adoption among smallholder farmers is not a lack of features but a mismatch between interface complexity and the digital literacy, device capabilities and connectivity conditions of the target user base (Muromba *et al.*, 2025; Das *et al.*, 2025; KIPPRA, 2023).

UCD in the AgroNexus project operates through three mechanisms. First, the functional requirements in Section 3.2 were derived not only from the problem analysis in Chapter 1 but also from the specific user needs documented in the literature, including the three things farmers consistently said they needed from a digital platform: a reliable way to find buyers, a way to arrange delivery, and some indication of what demand will look like in the near future (Abate *et al.*, 2023). Second, the interface design principles described in Section 3.5 are grounded in research on what works for smallholder farmers in low-resource environments, including the four-step task completion limit, the three-colour status system and the use of visual communication alongside text (KIPPRA, 2023; Muromba *et al.*, 2025). Third, each iteration includes a review stage in which the interface is evaluated against the usability criteria derived from the literature before the next iteration begins.

3.1.3 Development Phases

The combination of the Agile-Iterative methodology and User-Centred Design principles produces a development process organised into five phases, each of which is executed iteratively rather than as a single pass. The phases are requirements analysis, system design, implementation, testing and deployment.

During the requirements analysis phase, the coordination failures identified in Chapter 1 and the lessons drawn from the literature in Chapter 2 were translated into the functional and non-functional requirements presented in Section 3.2. In the system design phase, the architecture, database schema, AI model design and interface layout were specified in sufficient detail to begin implementation. The implementation phase covers the construction of the React.js web application, the Flutter mobile application, the Node.js backend and the Python forecasting microservice, described in Sections 3.5 and 3.6. The testing phase includes unit testing of individual components, integration testing of cross-component workflows such as the order-to-transport-request chain, and usability evaluation of the interface against the design principles. The deployment phase covers the configuration of the production environment and the staged release of the platform.

Figure 3.1 illustrates the iterative development cycle adopted for the AgroNexus project, showing how each phase feeds back into the previous phases as understanding deepens through successive iterations.

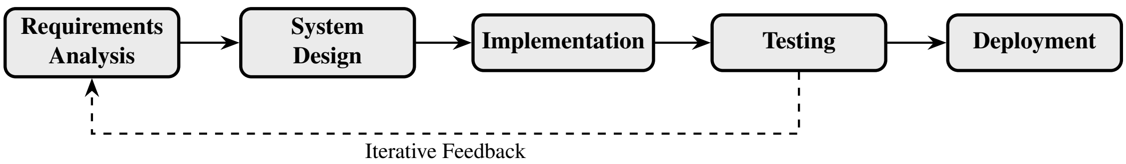

Figure 3.1: Agile-Iterative Development Cycle Adopted for AgroNexus

3.2 System Requirements

The requirements for AgroNexus were derived through the requirements analysis phase of the Agile-Iterative methodology described in Section 3.1. The coordination failures identified in Chapter 1 and the lessons drawn from the literature in Chapter 2 were translated into two categories. Functional requirements define what the system must do; non-functional requirements set the quality standards it must meet across performance, security, usability and reliability.

3.2.1 Functional Requirements

For AgroNexus, the functional requirements are organised by user role, since a farmer, a consumer and a transporter each need a different set of capabilities from the same underlying system.

i. *User Registration and Authentication:* The system shall support account creation and login for farmers, consumers and transporters. Each user shall only have access to the screens and features relevant to their assigned role. Access shall be enforced using JSON Web Tokens (JWT), with each token encoding the user's identity and role and being required on every API request.

ii. *Produce Listing and Management:* Registered farmers shall be able to create produce listings specifying crop type, available quantity, price per unit, farm location and the date from which the produce is available. Farmers shall be able to update quantities and prices as their stock changes and remove listings when produce is no longer available.

iii. *Browse and Order Placement:* Consumers shall be able to browse all active produce listings on the platform and filter them by crop type, region, price range and availability date. Any consumer who selects a listing shall be able to place an order directly, specifying the quantity required. The order shall be recorded in the system and the relevant farmer shall be notified.

iv. *Demand Forecasting:* The system shall maintain a machine learning module that processes historical order records, market prices, crop types, regional identifiers and seasonal variables to generate a rolling seven-day demand forecast for each crop category by region. The model shall be retrained periodically as new transaction data accumulates. Forecasts shall be displayed on the farmer dashboard to support planting and harvest timing decisions.

v. *Real-Time Market Dashboard:* The platform shall provide a dashboard accessible to all authenticated users showing current market prices by crop type and region, supply and demand levels, recent order activity and forecast trends. The dashboard shall update automatically whenever new listings, orders or price records are added to the database.

vi. *Transport Coordination:* On receipt of a new order, the system shall create a transport request and make it visible to registered transporters operating in the relevant region. The request shall display the pickup location, crop type, quantity and required delivery date. A transporter shall be able to accept the request through the platform, at which point both the farmer and consumer shall be notified.

vii. *Order Tracking and Notifications:* The status of each order shall be visible to all parties involved and shall update in real time as the order progresses through placement, acceptance, collection and delivery. Notifications shall be delivered within the application. Where a user's data connection is unreliable, SMS shall be used as a fallback channel.

viii. *Offline Access:* The Flutter mobile application shall cache key data on the device so that users can browse listings, review demand forecasts and check order status without an active data connection. Any changes made while offline shall be queued and submitted to the server automatically when the connection is restored.

3.2.2 Non-Functional Requirements

Non-functional requirements define the quality standards the system must meet across performance, security, usability and reliability.

i. *Performance:* Standard read requests to the API shall return a response within two seconds under typical usage load. Demand forecast generation shall complete within five seconds of a request.

ii. *Scalability:* Each major component of the backend, including authentication, listings, orders, transport and forecasting, shall be independently deployable so that any single component can be scaled up to handle increased load without requiring changes to the rest of the system.

iii. *Usability:* A user shall be able to complete their primary task, whether creating a listing, placing an order or accepting a transport request, within four steps of logging in. The interface shall undergo usability testing with representative users from each role before deployment.

iv. *Reliability:* The web application and API shall maintain availability of at least 99 percent. The mobile application shall operate in a read-only offline mode if the data connection is lost, without data loss or corruption when connectivity is restored.

v. *Security:* User passwords shall be stored as salted cryptographic hashes. All API endpoints shall require a valid JWT. All communication between client applications and the server shall use HTTPS. Role-based access controls shall prevent users from reading or modifying records that belong to other users or other roles.

vi. *Device Accessibility:* The Flutter mobile application shall function correctly on entry-level Android smartphones available in Ghana, including devices up to five years old with 2GB of RAM and slower processors.

3.2.3 Block Diagram

Figure 3.2 provides a high-level view of the AgroNexus system, showing the main components and the relationships between them.

The user interface layer, comprising the web and mobile clients, communicates exclusively with the API gateway, which routes requests to the business logic services and the database. The AI and machine learning engine sits alongside the API gateway as a separate service, callable on demand without being embedded in the main application logic. This separation ensures that any component can be updated independently without disrupting the rest of the system.

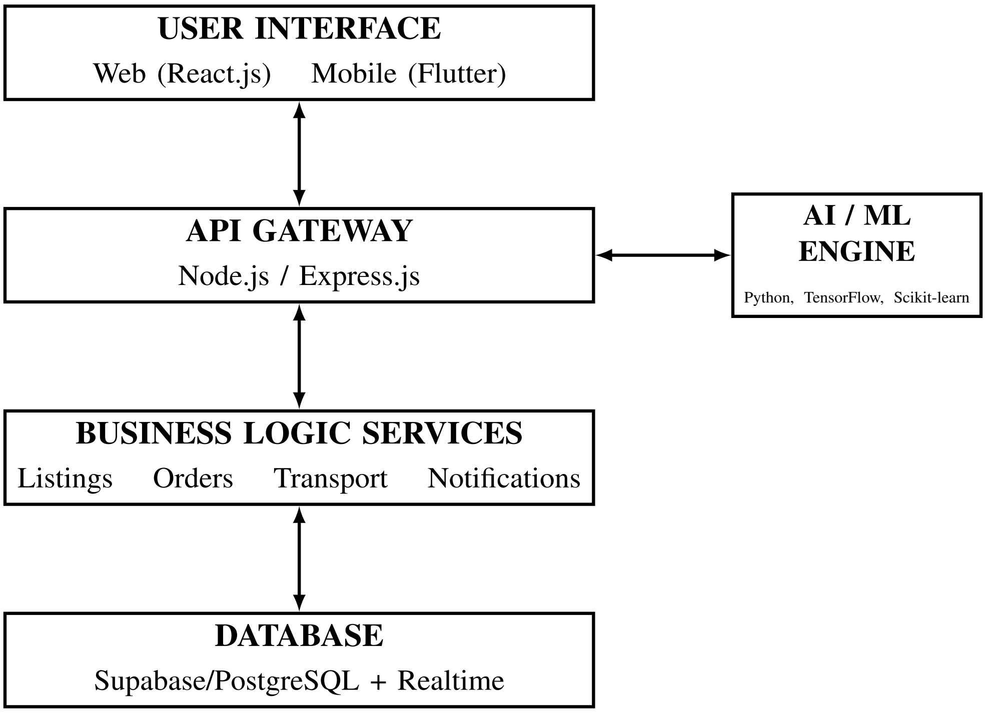

Figure 3.2: Block Diagram of Proposed System

3.3 System Architecture and Components

This section describes the system architecture, the data collection and model development pipeline for the demand forecasting function, the AI model design, and the data integration strategy that ties all components together.

3.3.1 System Architecture

AgroNexus follows a three-tier architecture separating the presentation, application logic and data layers to allow independent development, testing and scaling of each tier (Bass *et al.*, 2022). This architectural pattern isolates user-facing components from business logic and data storage, enabling each tier to be modified or replaced without affecting the others.

The presentation layer comprises a React.js web application and a Flutter mobile application. Both clients communicate with the backend through the same HTTP API endpoints.

The application logic layer is built on Node.js with Express.js. It handles authentication, listing management, order processing, transport coordination, dashboard aggregation and forecast routing. The demand forecasting model runs as a separate Python Flask microservice, callable via HTTP, enabling independent model updates without disrupting the main application (Richards and Ford, 2022).

The data layer uses Supabase (PostgreSQL) for relational storage, row-level security, authentication, real-time subscriptions and file storage. Real-time subscriptions push listing and order updates to all connected clients without polling.

Figure 3.3 shows the system architecture of the proposed AgroNexus model.

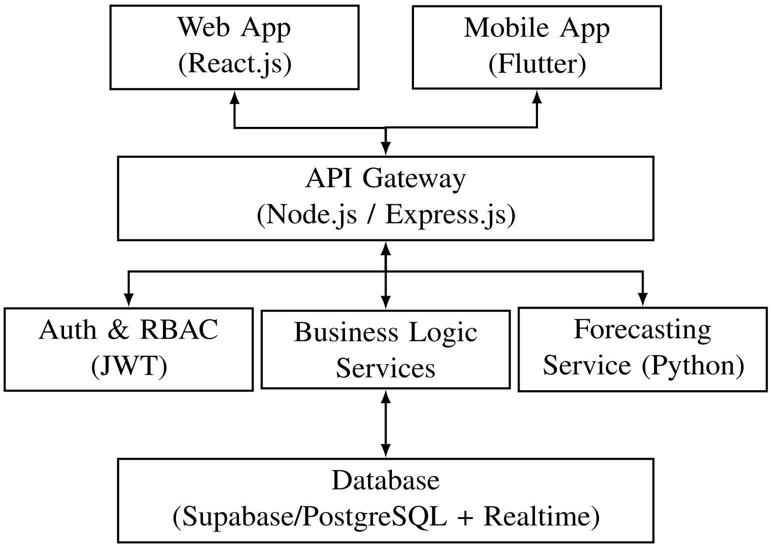

Figure 3.3: System Architecture of the Proposed Model

3.3.2 Data Processing and Model Development

Granular, transaction-level demand data for agricultural markets in the Western Region of Ghana is not publicly available. This limitation is acknowledged by the Ghana Ministry of Food and Agriculture's Statistics, Research and Information Directorate (SRID), which collects commodity inflow data administratively but does not publish it at market level. The training dataset for AgroNexus is therefore assembled from four complementary sources, each contributing a distinct layer of information that together provide a richer input than any single source alone could offer.

Primary Real Data: WFP Food Prices for Ghana

The primary source of real price data is the World Food Programme Food Prices dataset for Ghana, published and maintained by the WFP Vulnerability Analysis and Mapping unit and freely available for download through the Humanitarian Data Exchange platform (WFP, 2025). The dataset is updated monthly and contains market-level price records for major food commodities including maize, rice, cassava, plantain, tomatoes and pepper, collected from markets across all regions of Ghana. Records span from January 2006 to the present, providing nearly two decades of price observations that capture multiple full agricultural seasonal cycles. This dataset is downloaded directly as a CSV file and requires no institutional request or special access. It constitutes the backbone of the model's price input features and the primary source of the seasonal patterns the LSTM is trained to recognise and project forward.

Secondary Real Data: MOFA Facts and Figures

The second source is the *Agriculture in Ghana: Facts and Figures* annual report published by the Statistics Research and Information Directorate of the Ministry of Food and Agriculture (MOFA, 2025). Published annually since 1991, the most recent edition covers data through 2024. The document includes annual nominal and real average rural wholesale prices per metric ton for all major crops by region, annual production volumes in metric tons per crop per district, and cropped area estimates across all seventeen agricultural districts of the Western Region. These figures are extracted manually from the report appendices and serve two purposes within the data pipeline. They provide an independent corroboration of the price trends observed in the WFP dataset, allowing cross-validation of price magnitudes across sources. More critically, the annual production volumes per crop per region serve as calibration anchors for the synthetic demand volume data described below, ensuring that constructed demand figures are proportional to what Ghana's agricultural sector actually produces rather than being arbitrarily generated.

Supplementary Real Data: Weather Variables

Agricultural demand in Ghana is driven significantly by the bimodal rainfall pattern of the Western Region, where a major season runs from March to July and a minor season runs from September to November. Demand for perishable crops rises sharply in the weeks immediately following harvest as supply peaks, and falls during off-season periods. To give the LSTM access to the climatic signal that underlies these patterns, daily weather data including rainfall, temperature and humidity is sourced from the NASA Prediction of Worldwide Energy Resources (POWER) database (Stackhouse *et al.*, 2018). NASA POWER provides freely downloadable daily meteorological records for any geographic coordinate from 1981 to the present. Records are extracted for the coordinates of Tarkwa in the Western Region, covering the same date range as the WFP price data. Monthly aggregates of rainfall and temperature are computed from the daily records and added as input features. This approach is consistent with Nebri *et al.* (2024), whose agricultural demand forecasting model for Morocco incorporated seasonal climate variables alongside price records and demonstrated measurable improvement in forecast accuracy as a result.

Synthetic Demand Volume Data

The fourth layer of the training dataset addresses the absence of publicly available commodity inflow and order volume records for Western Region markets. Following the precedent established in AI system development for data-poor environments, where Dhal and Kar (2024) recommend staged deployment beginning with simpler models and constructed data before transitioning to real transaction records, and where synthetic data generation is a standard methodology when primary data is unavailable due to privacy, access or infrastructure constraints, synthetic demand volume records are constructed to serve as the target variable for model training.

The synthetic records are not randomly generated. Each record is built up in three stages. A baseline annual demand volume per crop per market location is derived from the MOFA production statistics, applying a consumption rate assumption drawn from the MOFA food balance sheet (MOFA, 2025); this ensures that the absolute scale of synthetic demand is consistent with actual crop availability in the Western Region. Within-year variation is then applied using Ghana's published crop harvest calendar, so that demand peaks align with known harvest periods for each crop type. Short-term price responsiveness is imposed on top of this by inverting the WFP price signal: weeks of falling prices correspond to periods of supply surplus and therefore higher simulated demand, while weeks of rising prices correspond to tighter supply and therefore lower simulated demand. Festival and calendar event flags covering Christmas, Easter, Eid al-Fitr, Homowo and school term transitions are applied as binary features that shift demand upward during the two weeks surrounding each major event.

The result is a synthetic demand volume series anchored to real production data, consistent with actual price dynamics and structured around real seasonal and calendar patterns. It does not reflect actual transaction records, and the model trained on it will require retraining once real order data accumulates on the AgroNexus platform. This limitation is acknowledged and forms the basis for the staged model deployment strategy described in Section 3.2.3, in which the gradient boosting baseline serves as the primary production model during the early operational phase while the LSTM is positioned for post-deployment retraining on real transaction sequences.

Data Preparation Pipeline

Before training begins, all records from the four sources are merged into a unified dataset indexed by date, crop type and market location. The following preparation steps are applied. Price and synthetic volume values are scaled to a consistent numerical range using min-max normalisation so that the model assigns equal weight to features of different magnitudes. Crop type and market location are encoded as integer category codes. The month of the year, week of the year and day of the week are extracted from each record's timestamp and added as separate features, encoding the seasonal and weekly rhythm of agricultural demand at multiple temporal scales. Lagged price features are constructed at intervals of seven, fourteen and thirty days, giving the LSTM explicit access to recent price momentum rather than requiring it to infer momentum from the sequence alone. Records containing apparent data entry errors are identified using interquartile range analysis and removed prior to training.

The combined dataset, spanning the WFP price records from 2006 to the present with aligned weather, festival and synthetic volume features, yields an estimated fifteen to twenty thousand training records across six crop types and three Western Region market locations including Tarkwa, Bogoso and Prestea. This volume is sufficient to expose the LSTM to multiple full agricultural seasonal cycles and to support the eighty-twenty training and test split described in Section 3.2.3. The most recent twenty percent of records are reserved for the test set, ordered chronologically rather than selected randomly, to prevent the model from appearing to perform well simply by having seen data recorded after the data it is evaluated on.

Once the AgroNexus platform is operational and real order and transaction records begin to accumulate, the synthetic demand volume layer will be progressively replaced by real platform data. The preprocessing pipeline is designed to accept either source without modification to the model architecture, so retraining on real data requires no structural changes to the system.

3.3.3 AI Demand Forecasting Model

The AgroNexus demand forecasting module frames the prediction task as a time series forecasting problem (Hewamalage *et al.*, 2023). The data the platform collects, including daily order volumes, market prices, crop types and geographic regions over time, is sequential and time-ordered. Agricultural demand follows seasonal cycles, responds to price movements in predictable ways, and is shaped by recurring calendar events. A time series model is the natural fit for data with this kind of structure.

The primary model selected for this task is a Long Short-Term Memory (LSTM) network implemented in TensorFlow (Sherstinsky, 2022). An LSTM is a neural network that reads data in time order, one step at a time, while keeping track of information from earlier steps in what is called a cell state. Unlike a standard neural network that treats each input independently, an LSTM can use patterns from weeks or months ago to influence its prediction for the current week. For agricultural demand, this matters because a crop's sales in late harvest season reflect what happened earlier in the same season, and are shaped by what happened during the same period the previous year. A forecasting model that cannot look back across this kind of time span will miss the most important patterns in the data.

The LSTM will be trained on thirty-day sequences of historical order volume and market price data for each crop type and region combination. The target output for each sequence is the total expected order volume over the following seven days. Training will use an eighty-twenty split, with the most recent twenty percent of records reserved for the test set, ordered chronologically to prevent data leakage. Model performance will be measured using Mean Absolute Percentage Error (MAPE) and Root Mean Square Error

(RMSE), with a target MAPE of twenty percent or below, corresponding to the eighty percent accuracy objective stated in Chapter 1.

During the early phase of development, before sufficient transaction data has accumulated to train the LSTM effectively, a gradient boosting model implemented in Scikit-learn will serve as a simpler baseline (Bentéjac *et al.*, 2021). Gradient boosting can produce useful predictions with smaller datasets than deep learning models require, and its output is easier to interpret and validate. The two models will be evaluated on the same test set and compared directly. As more data accumulates on the platform, the LSTM is expected to outperform the gradient boosting model, at which point it will be promoted to the primary production model.

Figure 3.4 illustrates the architecture of the LSTM forecasting model used in AgroNexus. The input layer receives a thirty-day sequence of features including price, volume, weather and calendar variables. Two stacked LSTM layers extract temporal patterns at different levels of abstraction. A dropout layer between them prevents overfitting. The final dense layer produces the seven-day demand forecast as a single output value.

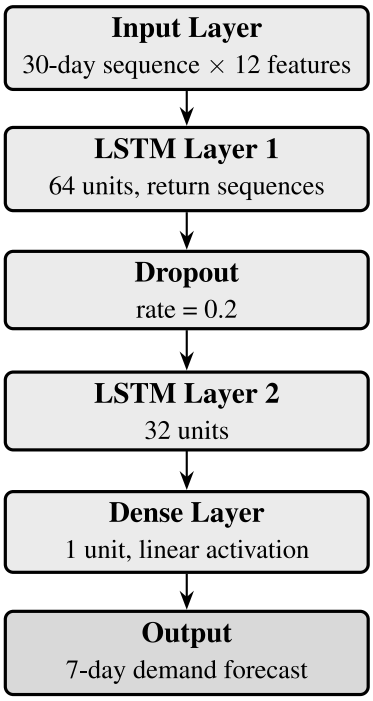

Figure 3.4: LSTM Demand Forecasting Model Architecture

3.3.4 Data Integration

All platform data, including listings, orders, prices, transport records and forecast output, is stored in a single Supabase database that every part of the system reads from. This shared data store is what makes the different components of AgroNexus work together rather than in isolation. The forecasting model reads from the orders and price records tables and writes its predictions back to a forecasts table. The dashboard reads from the listings, orders, prices and forecasts tables and assembles the view that users see. The transport coordination module reads from the orders table to generate requests and writes to the transport requests table when a transporter accepts. The Node.js API serves as the coordinator, directing requests from clients to the right service and ensuring that every write to the database follows the correct validation and authorisation checks.

3.4 Website Design

The AgroNexus web application delivers the full feature set of the platform through a browser interface accessible on any computer or tablet. This section describes the core features available to each user role and the interface design principles that guided layout and interaction patterns.

3.4.1 Core Features

The core features are organised by user role. The **Farmer Dashboard** displays active listings, current order counts, recent market prices, pending orders and seven-day LSTM demand forecasts per crop type. The **Consumer Browse** page presents a searchable, filterable list of active produce listings with crop type, region, price range and availability date filters. The **Transporter Feed** shows open delivery requests within the transporter's operating area, including pickup location, crop type, quantity and distance. The **Market Dashboard**, accessible to all roles, shows commodity price trends, regional supply-demand indicators and recent trading activity.

3.4.2 User Interface Design

Four principles drawn from the literature reviewed in Chapter 2 guide the interface design.

The first principle is role clarity. When a user logs in, they arrive at a screen that is specific to their role and contains only the features they need. A farmer does not see the consumer browse page or the transporter feed. A transporter does not see the produce listing creation form. Every user goes directly to the tools relevant to them. This role-specific approach follows the design pattern demonstrated by KhetBazaar, which showed that organising features by user role simplifies onboarding and reduces confusion (Maatman and Rosenstock, 2024).

The second principle is simplicity of workflow. The most important task for each role, whether creating a listing, placing an order, or accepting a transport request, must be completable within four steps from the home screen. This limit is not arbitrary. Research shows that smallholder farmers in low-literacy contexts abandon digital tools when a task requires more steps than they can track from memory (KIPPRA, 2023).

The third principle is visual communication. Status information is shown using colour and iconography as well as text. Order status uses a three-colour system: red for pending, amber for in progress and green for completed. Demand forecast levels are shown with directional arrows and shading rather than only as numbers. This ensures that users who find reading difficult can still understand the key information on the screen at a glance (Muromba *et al.*, 2025).

The fourth principle is offline resilience. When the mobile application is operating on cached data, a clearly visible indicator is shown on the screen so that the user knows the information may not be current. Write operations such as new listings or order placements are queued locally and the user is shown their queued status, so they know the action has been recorded and will be submitted when connectivity returns (Muromba *et al.*, 2025; Das *et al.*, 2025).

3.4.3 Technology Adoption Framework

Table 3.1 brings together the ten design decisions made across Sections 3.1 to 3.3 and organises them into three interdependent layers. Each layer addresses a different stage of the adoption journey: the foundation layer defines the constraints of the target environment; the first layer ensures the platform is accessible despite those constraints; the second layer builds user confidence through reliability and transparency; and the third layer keeps the platform financially and technically viable over time. Together, the ten drivers form a mutually reinforcing system in which no single feature works in isolation.

Table 3.1: Three-Layer Technology Adoption Framework for AgroNexus

Layer	**Focus**	**Adoption Drivers**
Operational Viability	Keeps the platform sustainable over time	Coordinator-not-owner model with per-transaction cost structure; staged AI deployment using gradient boosting baseline transitioning to LSTM; real-time dashboard updates via Supabase subscriptions eliminating manual refresh

Layer	**Focus**	**Adoption Drivers**
Trust and Reliability	Builds user confidence in the platform	Transparent real-time dashboard showing current prices, supply and demand levels, and seven-day AI forecast; SMS fallback for critical notifications when data connection is unavailable; integrated transport coordination where buyer and carrier are arranged through the same interaction
User-Centric Design	Enables access despite infrastructure constraints	Offline-first functionality with cached data, queued writes and automatic sync; low-device optimisation through Flutter native compilation and entry-level Android support; role-based and visual interface with primary task completable in four steps or fewer using colour and icon status indicators
Ghana Rural Context	Target environment defining all design constraints	Patchy 2G/3G connectivity; entry-level Android handsets two to five years old with 2 GB RAM; varying levels of digital and English literacy; seasonal income and transaction volumes

The layers are ordered from the most visible to the user (Layer 1) to the most structural (Layer 3). A platform that fails at Layer 1 will never reach adoption regardless of how well it performs at Layers 2 and 3. Equally, a platform that achieves wide adoption but fails at Layer 3 will eventually withdraw service, as the Twiga Foods case demonstrated. AgroNexus is designed to be sound at all three layers simultaneously.

3.5 Website Development

This section covers the implementation of the frontend and backend components of AgroNexus. The frontend encompasses the React.js web application and the Flutter mobile application. The backend covers the Node.js and Express.js API server, the JWT authentication system and the Python Flask forecasting microservice.

3.5.1 Frontend

The React.js web frontend uses a component-based structure with three protected route groups for the three user roles. An authentication guard checks the JWT on every page load and redirects unauthenticated users to the login screen.

The Flutter mobile frontend compiles from a single codebase to native Android and iOS. It shares the same API calls as the web application through a common API client layer that handles JWT attachment, request retry and error reporting. Offline caching uses Flutter's Hive library for local storage. A connectivity monitor triggers data synchronisation when the device transitions from offline to online.

3.5.2 Backend

The Node.js backend is structured as feature modules (authentication, profiles, listings, orders, transport, dashboard, forecasting), each following a three-layer pattern: route handler, service layer and data access layer.

For authentication, a JWT signed with a private key is issued on login, encoding the user's ID, role and expiry time. Every API request must include this token. The role claim enforces role-based access control at each endpoint.

The Python Flask forecasting service runs independently. The Node.js application sends HTTP POST requests with crop type, region and horizon parameters; Flask loads the trained model, runs the prediction and returns JSON. If the service is unavailable, cached forecasts are returned from the database.

The Supabase database contains five main tables: *users* (account and role data), *produce_listings* (crop type, quantity, price, location), *orders* (linked to listings), *transport_requests* (delivery status) and *price_records* (historical prices by crop, region and date).

3.6 Data Flow

This section traces how data moves through AgroNexus and shows how the different components work together.

When a farmer creates a listing, the app sends the details to the Node.js API, which validates the input, writes to the listings table and returns a success response. Supabase's real-time subscription pushes the new listing to any consumer with the browse page open.

When a consumer places an order, the API creates an order record, decrements the listing quantity, creates a transport request and notifies the farmer. The request is immediately visible on the transporter feed.

When a transporter accepts a request, the API updates the request status and notifies both the farmer and consumer.

The forecasting service runs nightly, reading thirty days of order and price records, processing them through the LSTM model and writing seven-day demand forecasts to the database. Updated forecasts are served to farmer dashboards on next login.

Figure 3.5 illustrates the overall workflow of the AgroNexus system.

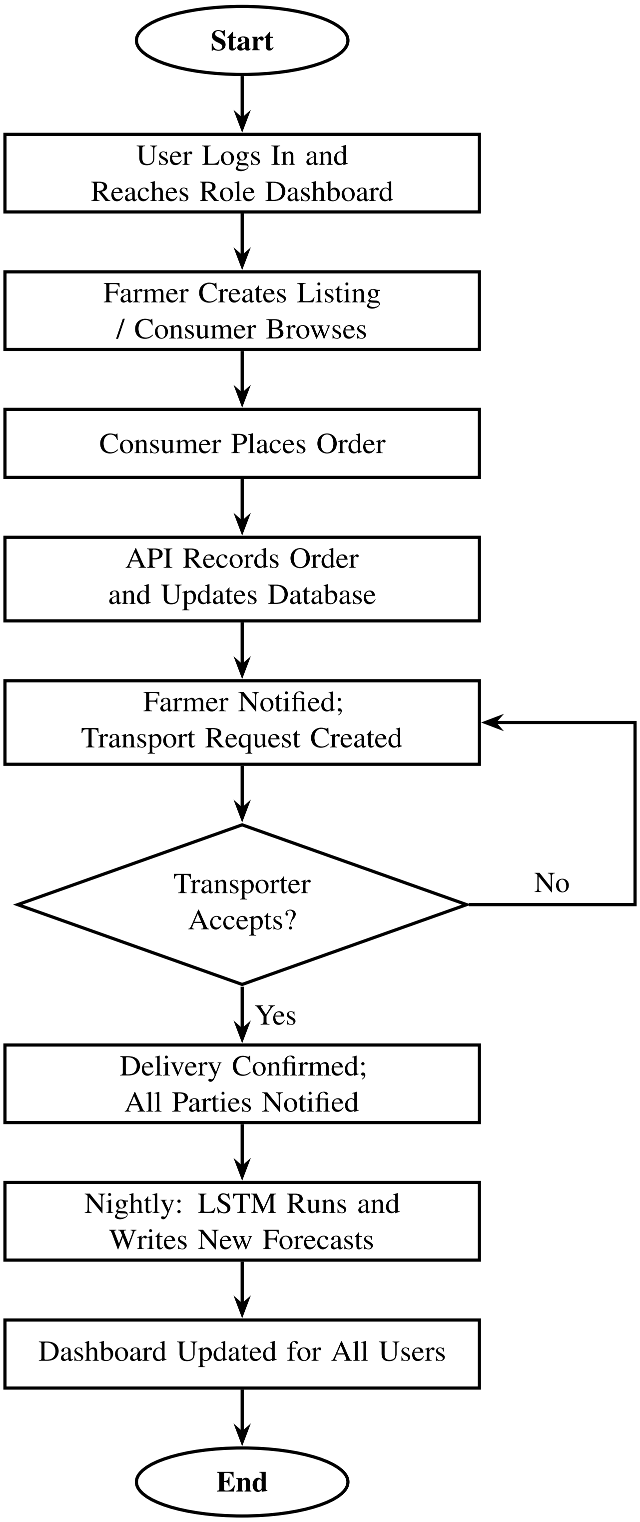

Figure 3.5: AgroNexus System Workflow

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